Rural Hospital Risk Indicators

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Introduction

Motivation

Rural hospitals are often the only source of care in their communities; financial and structural pressures have led to a steady number of hospital closures.

130+ Rural Hospitals Closed since 2013

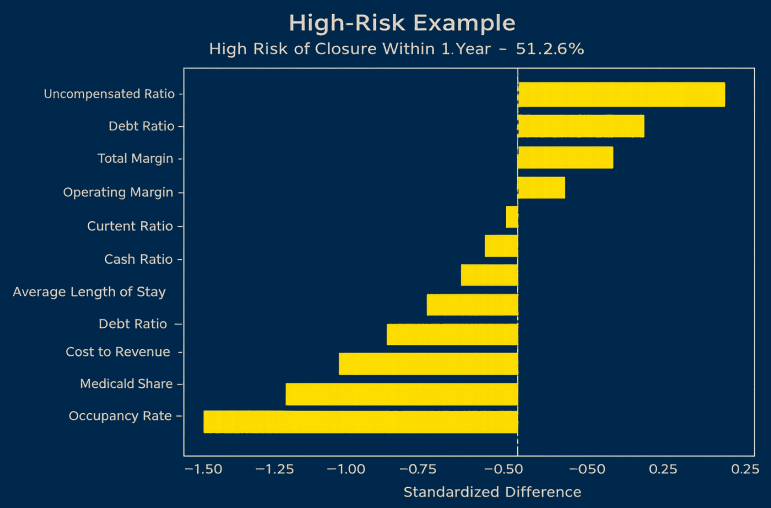
Goal

Develop an early warning system that identifies hospitals at risk of closure within the next 1-2 years. The goal is not perfect prediction but to highlight patterns of distress.

Data

This project uses national hospital-level data including financial, operational, and structural characteristics for years 2013-2023.

Top Drivers Relative to Average Hospital



Visual Insights

- High-risk example show lower occupancy and weaker margins
- Experiences higher costs relative to revenue
- Financial performance is below average across key metrics
- Large negative deviations signal elevated closure risk

Results

Key Drivers of Risk

High-risk hospitals show lower liquidity, weaker margins, higher costs, and greater reliance on public reimbursement.

Model Performance

- Identifying at-risk hospitals (recall)
- Limiting false alarms (precision)
- Balancing actionable thresholds

Conclusion

Clear, interpretable financial patterns are associated with hospital distress. Early identification supports more proactive decision making.

Impact

Helps policymakers and healthcare systems prioritize intervention before closure occurs.